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1. INTRODUCTION

1.1 Background

Multimodal knowledge discovery is the process of extracting and connecting useful information from different forms of data such as text, images, and audio-visual dialogue. In modern learning and information environments, a topic may be distributed across written notes, diagrams, lecture slides, photographs, and spoken explanations. Processing these sources separately can leave important relationships undiscovered. Multimodal knowledge discovery addresses this limitation by combining information from different modalities and representing related entities, concepts, events, and evidence in a structured form. Knowledge graphs provide a suitable representation because they can connect entities with relationships while also retaining supporting information such as source, timestamp, and evidence.

1.2 Problem Statement

The central problem addressed in this seminar is the development of a unified approach for discovering semantically connected knowledge from text, images, and audio-visual dialogue. Existing systems often process each modality independently, making it difficult to determine whether information from different sources refers to the same entity, concept, or event. Additional challenges include preserving temporal context, resolving duplicate or conflicting evidence, and maintaining provenance so that discovered facts can be verified. Therefore, the proposed approach focuses on cross-modal alignment, confidence-aware evidence fusion, and knowledge graph construction to create reliable and traceable multimodal knowledge.

1.3 Motivation

The motivation for this study comes from real-world situations where important information is distributed across multiple modalities. For example, a professor may explain a concept verbally, a lecture slide may provide its visual representation, and written notes may contain its definition. A system that processes these sources independently may produce disconnected information, whereas a multimodal knowledge discovery system can connect the related evidence and represent it as structured knowledge. Such an approach can be useful in educational content, technical

documentation, research assistance, multimedia archives, recorded lectures, and evidence-based information retrieval. The emphasis on provenance also allows users to trace discovered knowledge back to its original source.

1.4 Objectives

The main objectives of the proposed study are:

1. To process text, images, and audio-visual dialogue using modality-specific methods.
2. To extract entities, concepts, relations, events, visual objects, and speech segments.
3. To align semantically related evidence across different modalities.
4. To construct an evidence-grounded knowledge graph with source provenance.
5. To support query, retrieval, and explainable knowledge discovery.
6. To define an evaluation framework for extraction, alignment, retrieval, and graph quality.

2. LITERATURE REVIEW

2.1 Knowledge Graphs

Knowledge graphs organize information as entities and typed relationships. Ji et al. survey knowledge graph representation, acquisition, completion and applications, showing that graph construction depends on extracting structured facts from heterogeneous sources and that representation learning can support downstream reasoning and completion [1]. The graph abstraction is important for the proposed system because it provides an intermediate layer between raw multimodal observations and user-facing knowledge.

2.2 Multimodal Knowledge Graphs

Multimodal knowledge graphs extend conventional graph structures by linking entities with additional modalities such as images and text. Zhu et al. survey multimodal knowledge graph construction and applications and identify challenges in combining symbolic structure with multimodal evidence [2]. A later survey further categorizes construction, completion and applications, emphasizing named entity recognition, relation extraction, event extraction, entity linking and multimodal representation

learning [3]. These findings motivate a pipeline in which extraction and alignment are explicit stages rather than hidden inside one generative model.

2.3 Text and Document Understanding

Text processing commonly includes tokenization, sentence segmentation, named entity recognition, relation extraction and event extraction. For visually rich documents, the distinction between text content and document layout is also important. LayoutLM jointly models text and layout information from scanned document images and incorporates visual information for document understanding tasks [4]. Such work supports the use of OCR plus layout-aware processing for scanned pages and lecture slides.

2.4 Image and Vision-Language Understanding

Vision-language models learn relationships between visual content and language. BLIP provides a unified framework for vision-language understanding and generation and reports strong performance across image-text retrieval, captioning and visual question answering [5], [10], [11]. BLIP-2 further demonstrates an efficient strategy that connects frozen image encoders with large language models through a lightweight querying transformer [6]. For knowledge discovery, these approaches can be used as feature or evidence extractors, while the final structured representation remains a knowledge graph.

2.5 Speech and Audio-Visual Dialogue

Automatic speech recognition is a key bridge from audio to language. Whisper was trained on a large-scale collection of weakly supervised audio-transcript data and demonstrated robust multilingual speech recognition and zero-shot transfer [7], [13]. VideoBERT jointly models visual and linguistic sequences and shows that video-language learning can capture high-level semantic information for tasks such as action classification and video captioning [8]. These studies support extracting speech transcripts together with video timestamps and visual cues so that dialogue statements can be aligned with slides, objects or actions.

2.6 Multimodal Fusion

Multimodal fusion combines information from different sources into a representation that can support a downstream task. Recent surveys describe fusion as involving data representation, feature alignment, information integration and application-specific decision mechanisms [9], [14]. For knowledge discovery, fusion should not blindly average modalities; it should preserve evidence identity and allow confidence to depend on source quality, agreement and temporal or semantic alignment.

2.7 Research Gap

The literature demonstrates strong solutions for individual components, including knowledge graphs, document understanding, vision-language learning, speech recognition and multimodal fusion and evidence-grounded retrieval [15]. However, a student-oriented end-to-end architecture that explicitly connects text, images and audio-visual dialogue into one provenance-aware discovery pipeline remains a useful systems-level study. The proposed framework addresses this integration gap by treating each modality as an evidence source and using alignment, confidence and provenance as first-class components.

Work	Primary Focus	Relevance to Proposed System
Ji et al. [1]	Knowledge graph survey	Knowledge acquisition and representation
Zhu et al. [2]	MMKG construction and applications	Multimodal graph design
MMKG survey [3]	Construction, completion, applications	Cross-modal entity and relation handling
LayoutLM [4]	Text + layout + image	Scanned document understanding
BLIP [5]	Vision-language pretraining	Image-text evidence extraction
BLIP-2 [6]	Efficient vision-language learning	Visual semantic representation
Whisper [7]	Speech recognition	Audio-to-text evidence
VideoBERT [8]	Video-language learning	Audio-visual semantic alignment
Fusion survey [9]	Multimodal fusion	Evidence integration strategy

Table 2.1: Representative literature and contributions

3. SYSTEM ARCHITECTURE AND ANALYSIS

3.1 Existing Methodology

Existing systems commonly process one modality at a time. A text pipeline may identify entities and relations from documents, an image pipeline may perform OCR or object recognition, and a speech pipeline may convert audio into transcripts. The outputs are useful individually but are often stored in separate indexes. When a user asks a cross-modal question, the system must reconstruct the connection between sources at query time.

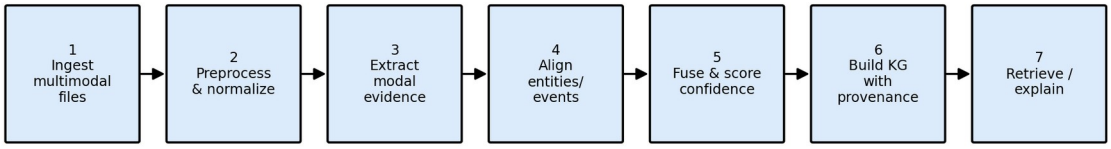


Figure 3.1: Existing sequential multimodal processing workflow

3.2 Limitations of Existing Approaches

The main limitation is the absence of a shared semantic layer. Text mentions may not be linked to corresponding image regions, and transcript sentences may not be connected to the visual frame or slide shown at the same time. Independent pipelines also make it difficult to resolve duplicate entities and to compare contradictory statements. Finally, if the final answer is generated without explicit provenance, users may not be able to trace the source of a discovered fact.

Aspect	Isolated Processing	Proposed Evidence-Grounded Approach
Representation	Separate modality outputs	Unified multimodal knowledge graph
Entity resolution	Mostly local	Cross-modal alignment
Temporal context	Often discarded	Timestamp-aware links
Evidence	Source may be hidden	Explicit provenance
Conflict handling	Limited	Confidence and agreement scoring
Querying	Modality-specific	Cross-modal graph retrieval

Table 3.1: Comparison of existing approaches

3.3 Functional Requirements

1. The system shall accept text, image and audio-visual inputs.
2. The system shall normalize and preprocess each modality.
3. The system shall extract entities, relations, events and modality-specific evidence.
4. The system shall align evidence referring to the same entity or event.
5. The system shall construct graph triples with source and confidence metadata.
6. The system shall support evidence-based querying and result explanation.

3.4 Non-Functional Requirements

- Accuracy: extracted knowledge should be evaluated using standard information extraction and retrieval metrics.
- Traceability: every important fact should have a source reference where possible.
- Scalability: the design should permit batch processing of many documents and recordings.
- Modularity: individual extractors should be replaceable without redesigning the whole system.
- Usability: users should be able to inspect the evidence supporting a discovered relation.
- Privacy: sensitive recordings and documents should be processed and stored with appropriate access controls.

3.5 Feasibility Analysis

The proposed architecture is technically feasible using open-source Python libraries and pretrained models. A prototype can begin with a small corpus and lightweight extractors before adding larger vision-language or speech models. The graph layer can be implemented with a property graph database or RDF store. The architecture therefore supports incremental development rather than requiring a single large end-to-end model.

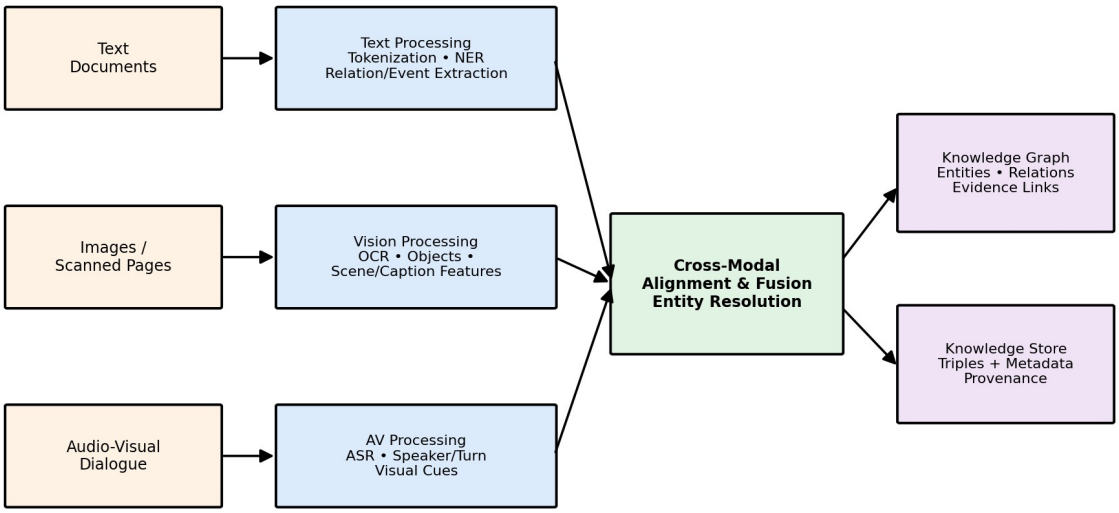


Figure 3.2: High-level analysis of the proposed multimodal discovery problem

4. SYSTEM DESIGN AND PROPOSED METHODOLOGY

4.1 Proposed Architecture

The proposed system consists of seven logical stages: ingestion, preprocessing, modality-specific extraction, cross-modal alignment, evidence fusion, knowledge graph construction and retrieval. The architecture deliberately separates extraction from knowledge representation. This makes it possible to inspect intermediate evidence and replace a model without changing the graph schema.

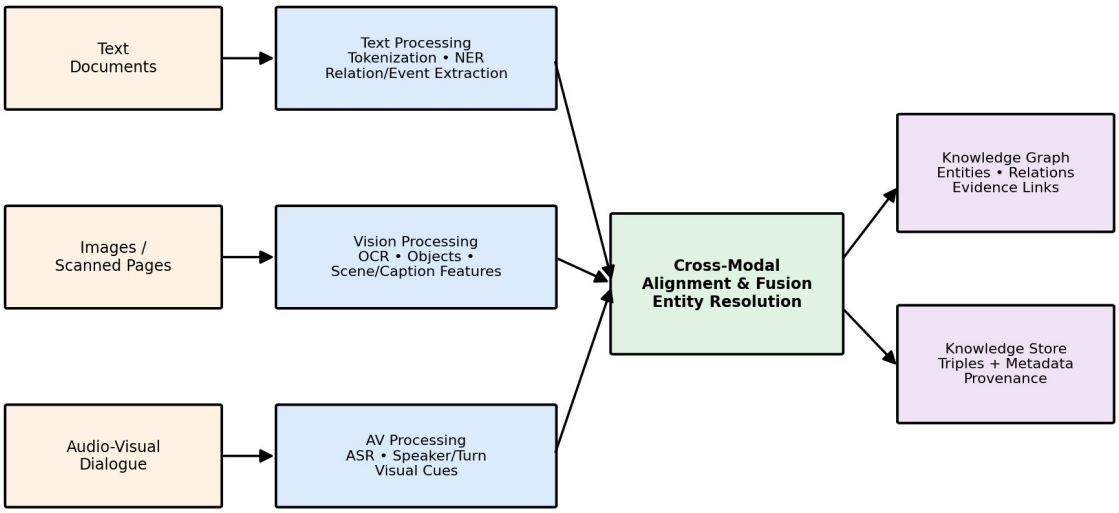


Figure 4.1: Proposed multimodal knowledge discovery architecture

4.2 End-to-End Pipeline

During ingestion, files are assigned identifiers and metadata. Preprocessing converts each source into machine-readable units such as text chunks, OCR tokens, image regions, transcript segments and video frames. Each extractor then produces candidate facts. Alignment determines whether two candidates refer to the same entity, event or concept. Fusion combines aligned evidence and assigns a confidence value. The resulting facts are stored as graph triples together with provenance.

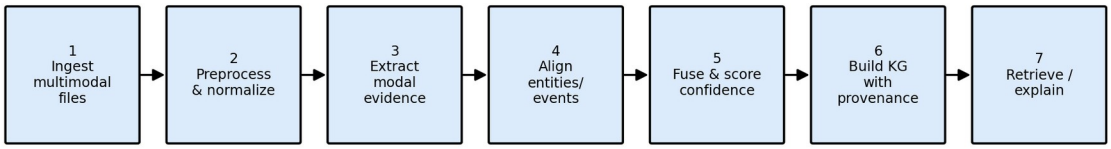


Figure 4.2: End-to-end knowledge discovery pipeline

4.3 Text Processing Module

The text branch begins with cleaning and segmentation. Named entity recognition identifies people, organizations, places, concepts and domain-specific terms. Relation extraction then identifies semantic links between entities, while event extraction captures actions and their participants. The extracted information is represented as candidate triples and linked to the original document span.

For example, a sentence such as “The professor explains convolution in a CNN lecture” can generate entities Professor, Convolution and CNN, together with relations such as teaches, explains or part-of depending on the domain ontology. The system should retain the sentence span rather than only the normalized triple so that the fact can be verified.

4.4 Image Processing Module

The image branch first determines whether the input is a scanned document, diagram, slide or natural image. OCR is applied when text is present. Layout information is retained for documents, while visual encoders or object detectors can provide semantic features for natural images and diagrams. Image regions are represented with coordinates so that extracted knowledge can point to a specific visual source.

4.5 Audio-Visual Dialogue Module

The audio-visual branch separates audio, speech and visual streams while retaining their common timeline. Automatic speech recognition converts speech into timestamped transcript segments. Speaker diarization can optionally identify speaker turns. Video frames are sampled around meaningful intervals, and visual features are

extracted from slides, objects, faces or scene changes. The transcript and visual evidence are then linked through timestamps.

4.6 Cross-Modal Alignment

Cross-modal alignment is the central step of the proposed system. An entity mentioned in text may appear in a slide, be spoken in a transcript and be visible in a diagram. Alignment combines lexical similarity, embedding similarity, metadata and temporal proximity. When the available evidence is insufficient, the system should preserve the candidates separately rather than forcing a false merge.

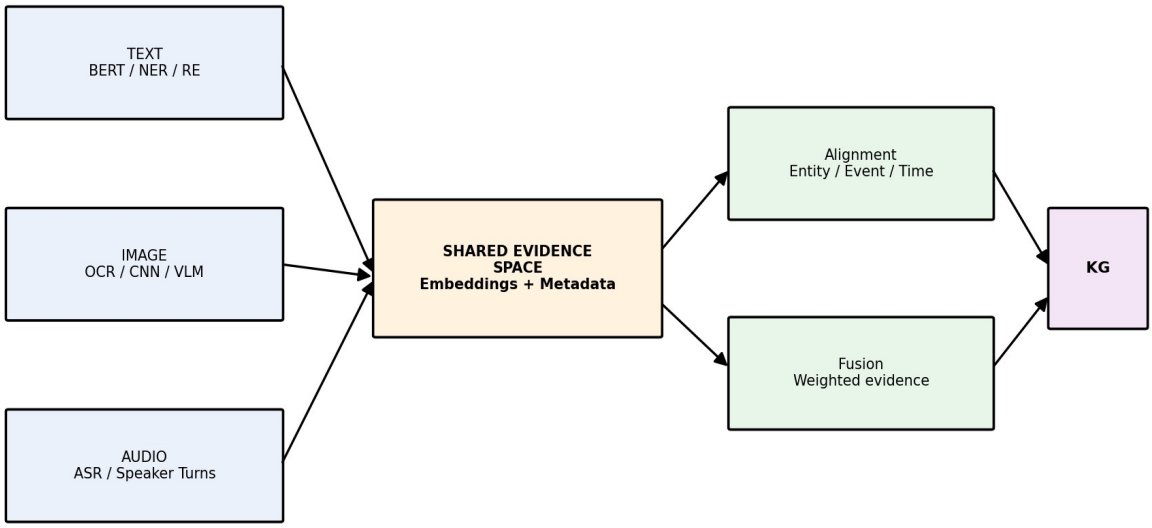


Figure 4.3: Cross-modal alignment and confidence-aware evidence fusion

4.7 Evidence Confidence Model

A simple confidence score can combine the reliability of each modality with the strength of cross-modal agreement. Let Ct, Ci and Ca denote confidence from text, image and audio-visual evidence, and let A denote the alignment score. A weighted model can be expressed as eqn. (4.1).

$$C = wt\ Ct + wi\ Ci + wa\ Ca + wf\ A \tag{4.1}$$

The weights should be selected according to the application. The score is not a claim of statistical probability; it is an engineering confidence value used to rank candidate facts and to decide when additional verification is necessary.

4.8 Knowledge Graph Model

The knowledge graph stores entities as nodes and semantic relationships as edges. Each fact can also store evidence identifiers, modality, timestamp, source document, confidence and processing model. This makes the graph evidence-grounded rather than a collection of unsupported statements.

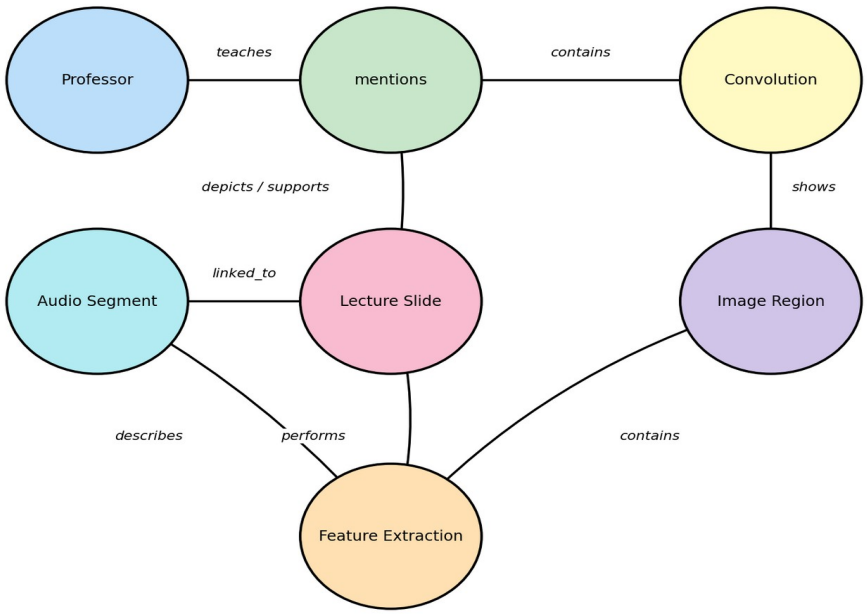


Figure 4.4: Example evidence-grounded knowledge graph for a CNN lecture

4.9 Proposed Data Schema

A consistent data schema is essential for storing extracted knowledge in a form that is both machine-queryable and human-verifiable. Each record in the proposed schema represents a single fact together with the metadata needed to trace it back to its origin. Separating the semantic content of a fact from its provenance and confidence allows the knowledge graph to be audited, corrected, and extended without redesigning the underlying storage. The schema below defines the fields used to represent each knowledge record.

Field	Description
Entity_ID	Unique identifier for an entity or concept
Entity_Type	Person, concept, object, organization, event, etc.
Relation	Typed semantic relationship

Field	Description
Source_ID	Document, image or recording identifier
Evidence_Span	Text span, image region or audio/video interval
Timestamp	Temporal location for dialogue/video evidence
Confidence	Engineering confidence score
Provenance	Model and source information used to derive the fact

Table 4.1: Proposed knowledge record schema

4.10 Query and Retrieval Design

A query is first converted into a structured intent containing entities, relations, time constraints and optional modality preferences. Graph traversal retrieves candidate facts, while evidence ranking prioritizes high-confidence and mutually supported facts. The final response should show the discovered relation together with concise evidence references, allowing the user to inspect why the relation was included.

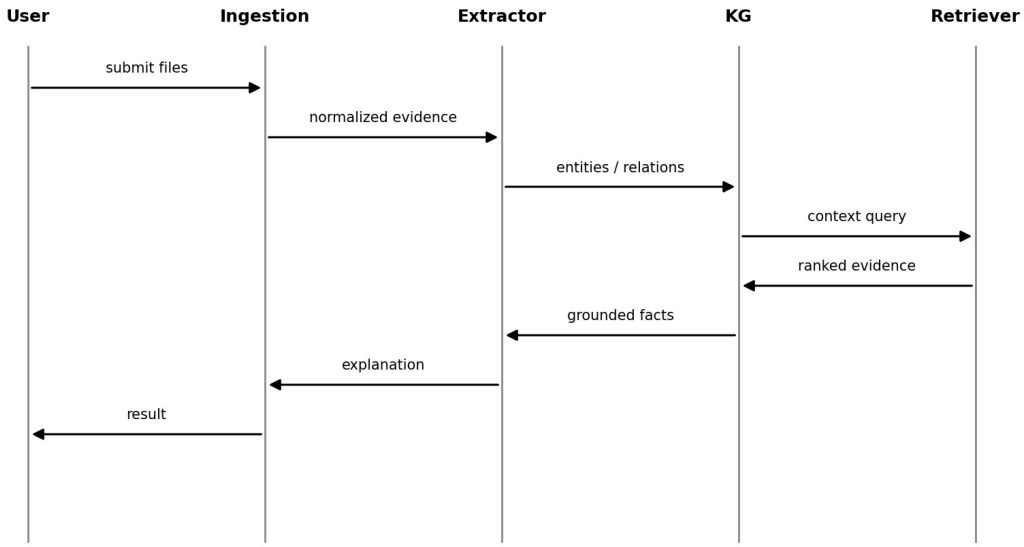


Figure 4.5: Query-to-evidence interaction sequence

4.11 Knowledge Discovery Algorithm

1. Read and register the available multimodal sources.

2. Normalize text, images, audio and video into processable units.

3. Extract candidate entities, relations, events and modality-specific features.

4. Generate embeddings and metadata for candidate evidence.

5. Align candidates across modalities using semantic and temporal similarity.

6. Compute confidence and retain provenance for each accepted fact.

7. Insert entities and relations into the knowledge graph.

8. Retrieve and rank evidence for a user query.

9. Return a structured result with supporting evidence.

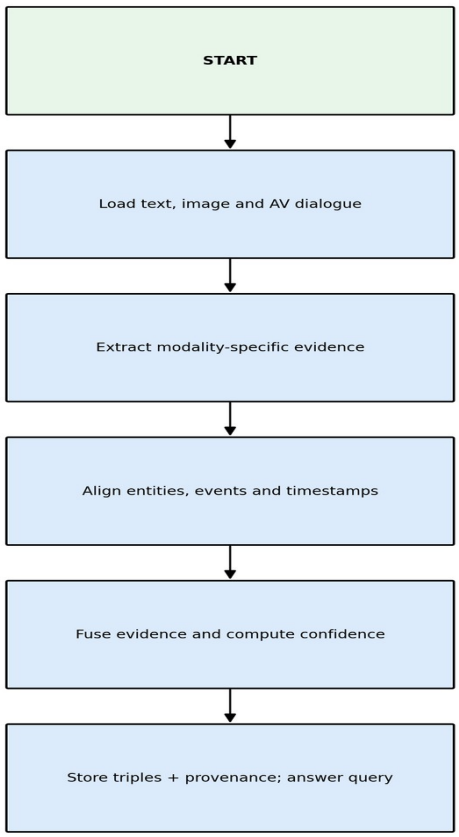


Figure 4.6: Knowledge discovery algorithm flow

5. IMPLEMENTATION DETAILS

5.1 Development Approach

The implementation can be developed as a modular Python-based application. Each modality has an independent preprocessing and extraction component, while a common evidence object provides a consistent interface to the alignment and graph layers. This architecture supports experimentation with different models without changing the overall data flow.

5.2 Suggested Technology Stack

The proposed system can be implemented entirely using mature, widely adopted open-source tools, which keeps the project feasible for a seminar-scale implementation while remaining extensible for future production use. Each layer of the stack is chosen to match the responsibility of a specific pipeline stage, from raw file ingestion through modality-specific extraction to graph storage and the user-facing API. The table below summarizes the suggested technology for each layer and its purpose within the overall architecture.

Layer	Suggested Technology	Purpose
Programming	Python	Pipeline orchestration and ML integration
NLP	spaCy / Transformers	NER, embeddings and relation processing
OCR	Tesseract / document OCR	Text extraction from images
Vision	OpenCV / pretrained vision models	Image and frame processing
Speech	Whisper or equivalent ASR	Timestamped speech transcription
Embeddings	Sentence-transformers / vision-language encoder	Semantic similarity
Graph	Neo4j or RDF store	Knowledge graph storage and traversal
API	FastAPI / Flask	Service layer
Frontend	HTML/CSS/JavaScript or React	Upload, query and evidence visualization
Storage	File store + metadata DB	Source and intermediate artifact management

Table 5.1: Suggested technology stack

5.3 Data Preparation

A prototype dataset should contain paired or related examples: a written explanation, a corresponding slide or diagram, and a recording containing the explanation. Each source should be assigned a unique identifier. Transcripts should retain timestamps, while images should retain file and region metadata. The dataset can be split into development, validation and test subsets if supervised evaluation is performed.

5.4 Processing and Storage

Processing can be implemented as a pipeline in which each stage writes structured intermediate objects. For example, the ASR stage produces transcript segments, the vision stage produces image observations, and the NLP stage produces entity and relation candidates. A common evidence schema allows these objects to be passed to the alignment stage. The graph database stores the final relationships, while large binary files remain in external storage.

5.5 Illustrative Deployment

A representative deployment separates the client-facing layer from the processing and storage layers so that each component can be scaled or replaced independently. The client submits files and queries through a web interface, the API layer routes requests to the appropriate processing or retrieval service, and the knowledge graph and vector store persist structured facts and embeddings respectively. All processing and retrieval activity can additionally be logged alongside supporting evidence so that discovered facts remain auditable after deployment.

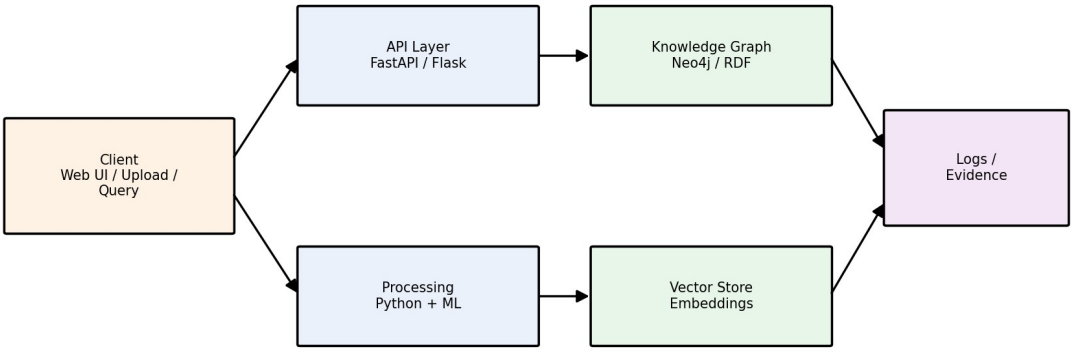


Figure 5.1: Illustrative deployment architecture

5.6 Evaluation Metrics

Since the proposed system combines several independent components, evaluation must be performed at both the component level and the system level. Component-level metrics assess the quality of individual extractors such as named entity recognition, OCR, and speech recognition, while alignment and retrieval metrics assess how well the system connects and ranks evidence across modalities. Using standard, well-understood metrics for each component allows the proposed architecture to be benchmarked against existing single-modality baselines. The following metrics are proposed for each component of the pipeline.

Component	Metric	Purpose
NER / extraction	Precision, Recall, F1	Measure correctness of extracted entities/facts
OCR	Character/word error rate	Measure text extraction quality
ASR	Word Error Rate (WER)	Measure speech transcription quality
Alignment	Precision, Recall, F1	Measure cross-modal entity/event matching
Retrieval	Precision@k, Recall@k, MRR	Measure evidence retrieval quality
Graph quality	Triple validity / coverage	Measure structural usefulness
Provenance	Evidence coverage	Measure traceability of facts

Table 5.2: Evaluation metrics

5.7 Illustrative Test Cases

To validate that the system behaves correctly across the range of situations it is designed to handle, a small set of illustrative test cases is proposed covering single-modality pairs, timestamp-based alignment, conflicting evidence, and cross-modal queries. These test cases are not exhaustive but represent the core scenarios that any evidence-grounded multimodal discovery system must handle correctly before it can be considered reliable.

Test Case	Input	Expected Discovery
T1	Text + lecture slide	Concepts in text linked to slide elements
T2	Lecture audio + slide	Transcript statement linked to slide timestamp
T3	Diagram + explanation	Visual component linked to spoken concept
T4	Conflicting sources	Lower-confidence or conflicting evidence retained
T5	Cross-modal query	Answer supported by multiple evidence sources

Table 5.3: Illustrative test cases

5.8 Expected Observations

Because this report describes a proposed seminar system rather than a completed experimental deployment, numerical performance results are not claimed. The expected benefit is qualitative: cross-modal alignment should improve the completeness and traceability of discovered knowledge compared with storing independent modality outputs. Evaluation should determine whether the added alignment complexity produces measurable gains in fact coverage, retrieval quality and evidence traceability.

5.9 Security and Ethical Considerations

Audio-visual dialogue can contain personal, confidential or sensitive information. The implementation should therefore use access control, secure storage, retention policies and consent-aware data collection. The system should distinguish extracted evidence from inferred relationships and should avoid presenting uncertain facts as verified truth. Human review is particularly important for high-impact domains.

5.10 Limitations

- Speech recognition may degrade with noise, accents, overlapping speakers or domain-specific vocabulary.
- OCR may fail on low-resolution images, handwriting, complex diagrams or unusual fonts.
- Entity alignment can produce false merges when similar names refer to different entities.
- Visual and audio streams may be poorly synchronized in real recordings.
- Confidence scores require calibration and should not be interpreted as absolute truth probabilities.
- Large multimodal models can increase computation, memory requirements and deployment cost.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

Knowledge discovery from text, images and audio-visual dialogue is an important systems problem because meaningful information is distributed across heterogeneous sources. The literature shows strong progress in knowledge graphs, multimodal knowledge graphs, document understanding, vision-language learning and speech recognition [1]–[9]. The proposed framework brings these capabilities together through a modular pipeline consisting of ingestion, preprocessing, modality-specific extraction, cross-modal alignment, confidence-aware fusion, knowledge graph construction and evidence-based retrieval.

The central design principle is evidence grounding. Instead of treating the output as an unsupported generated statement, each discovered relation is associated with source metadata and, where possible, an exact text span, image region or audio-video interval. This design can improve traceability and make the system more useful for educational content, technical documents, meeting analysis and multimedia archives.

6.2 Key Contributions of the Proposed Study

- Unified architecture for text, image and audio-visual dialogue knowledge discovery.
- Explicit cross-modal entity and event alignment layer.
- Confidence-aware evidence fusion rather than unqualified modality merging.
- Knowledge graph representation with provenance and temporal metadata.
- Query interface concept that returns evidence with discovered knowledge.
- Evaluation framework covering extraction, alignment, retrieval and traceability.

6.3 Future Scope

While the proposed architecture addresses the core integration problem, several extensions can further improve its accuracy, usability, and applicability. These enhancements span algorithmic improvements such as stronger multimodal models and dynamic graph updates, as well as practical concerns such as privacy-aware processing

and human-in-the-loop verification. The table below outlines the key directions for future work.

Area	Future Enhancement
Context awareness	Use conversation history, topic context and temporal context for better disambiguation.
Multimodal LLMs	Use stronger multimodal models as optional extractors while retaining graph grounding.
Dynamic graphs	Update knowledge continuously as new documents and recordings arrive.
Explainability	Generate human-readable evidence paths and confidence explanations.
Domain adaptation	Adapt the ontology and extractors for education, healthcare, engineering and legal archives.
Federated / privacy-aware processing	Support local processing where raw audio or documents cannot leave the source environment.
Benchmarking	Create paired text-image-video datasets with gold-standard cross-modal relations.
Human-in-the-loop	Allow reviewers to confirm, reject or correct uncertain graph relations.

Table 6.1: Future scope areas

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